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Sobel Edge Detection Accelerator on FPGA (PYNQ-Z2)

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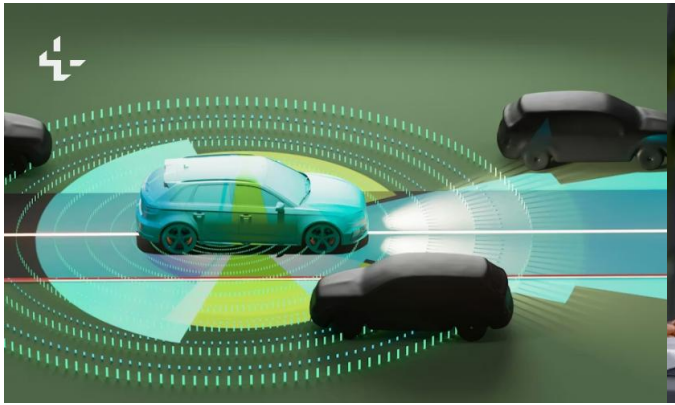
Background

- Edge detection is the first step in countless vision systems
- Identifies boundaries, shapes, and structures in images



Lane Detection

Object Recognition

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Project Overview

A **fully pipelined** Sobel edge detection accelerator synthesized with Vitis HLS and deployed on PYNQ_Z2, processing grayscale images via AXI_Stream at **one pixel per clock cycle** using BRAM line buffers and a register_partitioned 3×3 sliding window.

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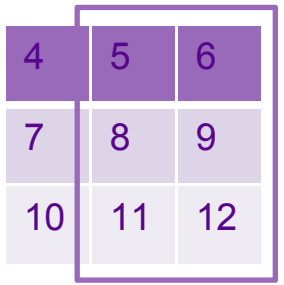
Sobel methodolodgy

Sobel Gradient Computation

- Sobel operator convolves a 3×3 kernel over each pixel to compute spatial gradients
- **Gx**: horizontal gradient kernel — detects vertical edges
- **Gy**: vertical gradient kernel — detects horizontal edges
- Gradient magnitude: **M = |Gx| + |Gy|**

generated window



3×3 Sliding Window

- A fully register-partitioned **3×3** window shifts one column per clock cycle
- Provides simultaneous access to all 9 neighboring pixels in a single cycle
- Enables **II=1 pipelined execution** — one output pixel per clock cycle

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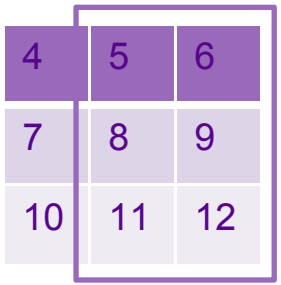
Sobel methodolodgy

Streaming Input — No Frame Buffer Required

- Image is received as a **pixel-sequential AXI-Stream**, one byte per transaction
- Full-frame storage is avoided — only **two BRAM line buffers** retained at any time
- Each line buffer stores one complete row (up to 1920 pixels × 8-bit)
- At any cycle: row N−2 and row N−1 are buffered; row N arrives live from the stream

generated window



5	6	line buffer(previous column)
8	9	line buffer(previous column)
11	12	new pixel

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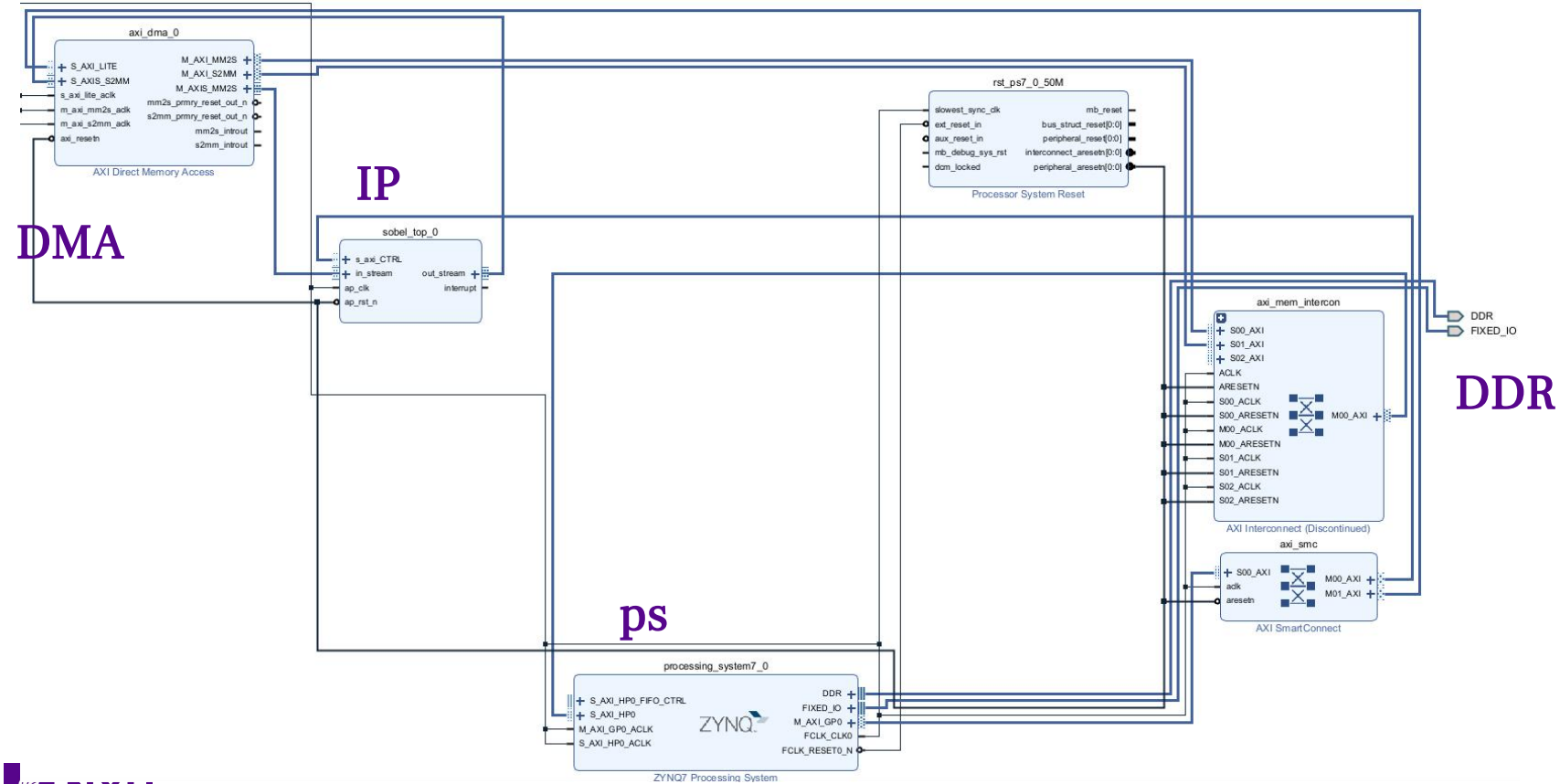
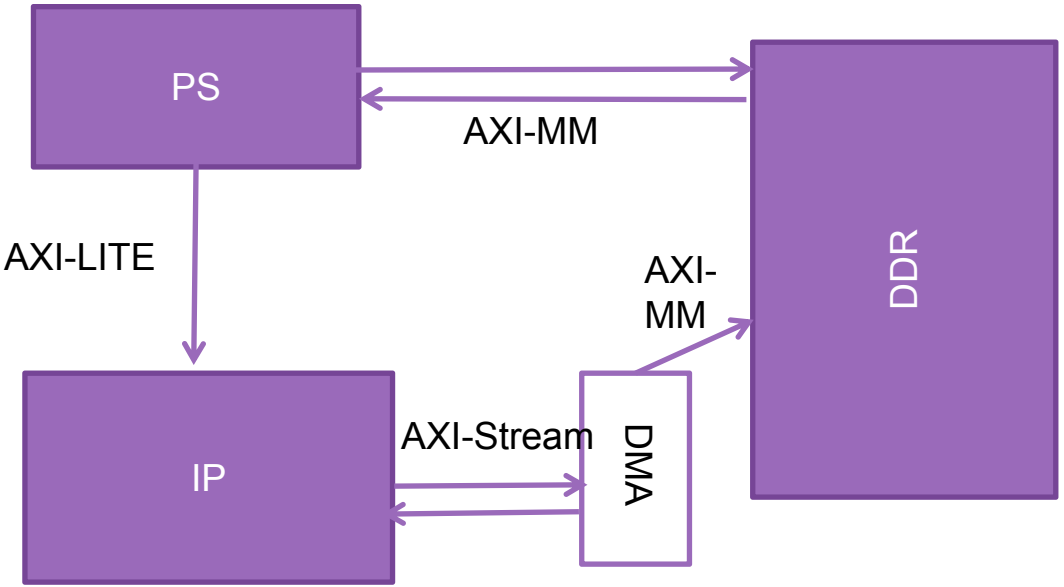

axi_dma_0
S_AXI_LITE
S_AXIS_S2MM
s_axi_lite_aclk
m_axi_mm2s_aclk
m_axi_s2mm_aclk
axi_resetn
M_AXI_MM2S
M_AXI_S2MM
M_AXIS_MM2S
mm2s_prmry_reset_out_n
s2mm_prmry_reset_out_n
mm2s_introut
s2mm_introut
AXI Direct Memory Access
DMA
IP
sobel_top_0
s_axi_CTRL
in_stream
ap_clk
ap_rst_n
out_stream
interrupt
rst_ps7_0_50M
slowest_sync_clk
ext_reset_in
aux_reset_in
mb_debug_sys_rst
dcm_locked
mb_reset
bus_struct_reset[0:0]
peripheral_reset[0:0]
interconnect_aresetn[0:0]
peripheral_aresetn[0:0]
Processor System Reset
axi_mem_intercon
S00_AXI
S01_AXI
S02_AXI
ACLK
ARESETN
S00_ACLK
S00_ARESETN
M00_ACLK
M00_ARESETN
S01_ACLK
S01_ARESETN
S02_ACLK
S02_ARESETN
M00_AXI
AXI Interconnect (Discontinued)
DDR
FIXED_IO
DDR
axi_smc
S00_AXI
aclk
aresetn
M00_AXI
M01_AXI
AXI SmartConnect
ps
processing_system7_0
S_AXI_HP0_FIFO_CTRL
S_AXI_HP0
M_AXI_GP0_ACLK
S_AXI_HP0_ACLK
ZYNQ
DDR
FIXED_IO
M_AXI_GP0
FCLK_CLK0
FCLK_RESET0_N
ZYNQ7 Processing System



Abstract graphic

AXI–Lite: used by the PS to configure the Sobel IP, including width, height, and control/status registers.

AXI–MM: used by the AXI DMA to read the input image from DDR and write the processed output image back to DDR.

AXI–Streaming: transfer the pixel data to IP FIFO/DMA FIFO, easy pipelining

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```
#include "sobel.hpp"        You, 2 weeks ago • main

void sobel_top(hls::stream<axis_t> &in_stream,
               hls::stream<axis_t> &out_stream,
               int width,
               int height) {
#pragma HLS INTERFACE axis port=in_stream
#pragma HLS INTERFACE axis port=out_stream

#pragma HLS INTERFACE s_axilite port=width  bundle=CTRL
#pragma HLS INTERFACE s_axilite port=height bundle=CTRL
#pragma HLS INTERFACE s_axilite port=return bundle=CTRL

    sobel_core(in_stream, out_stream, width, height);
}
```

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Storage definition

```
void sobel_core(hls::stream<axis_t> &in_stream,
                hls::stream<axis_t> &out_stream,
                int width,
                int height) {
#pragma HLS INLINE off

    if (width <= 0 || height <= 0) return;
    if (width > MAX_WIDTH || height > MAX_HEIGHT) return;

    ap_uint<8> linebuf0[MAX_WIDTH];//bram
    ap_uint<8> linebuf1[MAX_WIDTH];
      You, 2 weeks ago • main
#pragma HLS BIND_STORAGE variable=linebuf0 type=ram_2p impl=bram
#pragma HLS BIND_STORAGE variable=linebuf1 type=ram_2p impl=bram

    ap_uint<8> window[3][3];//partirion into register
#pragma HLS ARRAY_PARTITION variable=window complete dim=0
```

2Bram: Dual port support read and write simultaneously

9 registers

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3*3 window generate & linebuffer update

```
#include "sobel.hpp"        You, 2 weeks ago • main

void window_generator(ap_uint<8> pix,
                      ap_uint<11> col,
                      ap_uint<8> linebuf0[MAX_WIDTH],
                      ap_uint<8> linebuf1[MAX_WIDTH],
                      ap_uint<8> window[3][3]) {
#pragma HLS INLINE

    ap_uint<8> old0 = linebuf0[col];
    ap_uint<8> old1 = linebuf1[col];

    window[0][0] = window[0][1];
    window[0][1] = window[0][2];
    window[1][0] = window[1][1];
    window[1][1] = window[1][2];
    window[2][0] = window[2][1];
    window[2][1] = window[2][2];

    window[0][2] = old0;
    window[1][2] = old1;
    window[2][2] = pix;

    linebuf0[col] = old1;
    linebuf1[col] = pix;
}
```

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Pipeline and algorithm

```
row_loop:
    for (int row = 0; row < height; row++) {
col_loop:
        for (int col = 0; col < width; col++) {
#pragma HLS PIPELINE II=1

            axis_t in_pkt = in_stream.read();
            ap_uint<8> pix = in_pkt.data;

            ap_uint<11> col_u = col;
            window_generator(pix, col_u, linebuf0, linebuf1, window);

            ap_uint<8> out_pix = 0;

            if (row >= 2 && col >= 2) {
                ap_int<12> p00 = window[0][0];
                ap_int<12> p01 = window[0][1];
                ap_int<12> p02 = window[0][2];

                ap_int<12> p10 = window[1][0];
                ap_int<12> p12 = window[1][2];

                ap_int<12> p20 = window[2][0];
                ap_int<12> p21 = window[2][1];
                ap_int<12> p22 = window[2][2];

                ap_int<12> gx_left  = p00 + (p10 << 1) + p20;
                ap_int<12> gx_right = p02 + (p12 << 1) + p22;
                ap_int<12> gx = gx_right - gx_left;

                ap_int<12> gy_top = p00 + (p01 << 1) + p02;
                ap_int<12> gy_bot = p20 + (p21 << 1) + p22;
                ap_int<12> gy = gy_bot - gy_top;

                ap_uint<12> mag = abs12(gx) + abs12(gy);
                out_pix = clip_u8_12(mag);
            }
```

- Instead of buffering the full frame, only two preceding rows are stored in BRAM, while the active 3×3 neighborhood is held in fully partitioned registers for single-cycle parallel access.
- Applying pipeline to the outer row loop introduces loop-carried dependencies on line buffer updates, preventing II=1; the directive is therefore applied to the inner column loop, where all dependencies resolve within a single iteration.

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HLS Optimizations

#pragma HLS PIPELINE II=1

Applied to the inner column loop, enabling one pixel processed per clock cycle with full spatial reuse of line buffer data.

#pragma HLS ARRAY_PARTITION variable=window complete dim=0

Completely partitions the 3×3 window across all dimensions into dedicated registers, eliminating memory port contention and exposing all nine pixels to the datapath in a single cycle.

Bit-Width Optimization

Sobel gradients on 8-bit inputs require at most 12 bits. Explicit **ap_int<12>** specification avoids default 32-bit arithmetic, reducing LUT utilization and shortening the critical path.

Synthesis / Resource Results

Target clock: 8.00 ns / 125 MHz
Estimated clock: 5.799 ns / ~172 MHz
II: 1
BRAM_18K: 2
DSP: 1
FF: 1,197
LUT: 1,837
The target was 125 MHz. HLS estimated 5.799 ns, which is faster than the 8 ns target, so the design meets the HLS timing goal.

At II=1 and 125 MHz, the theoretical steady-state throughput is about 125 million pixels per second.

Resource usage is low because the design only keeps two line buffers and a 3×3 register window, instead of buffering the full frame.

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Estimated Quality of Results

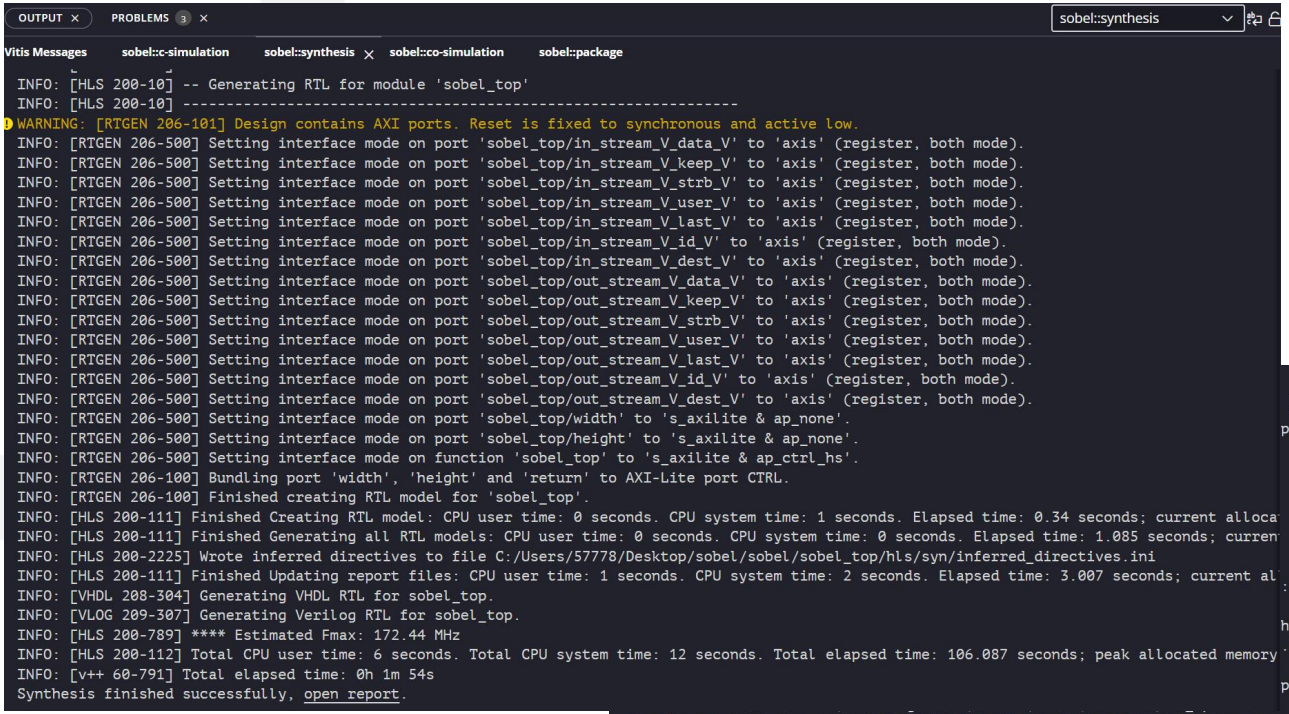
Timing Estimate

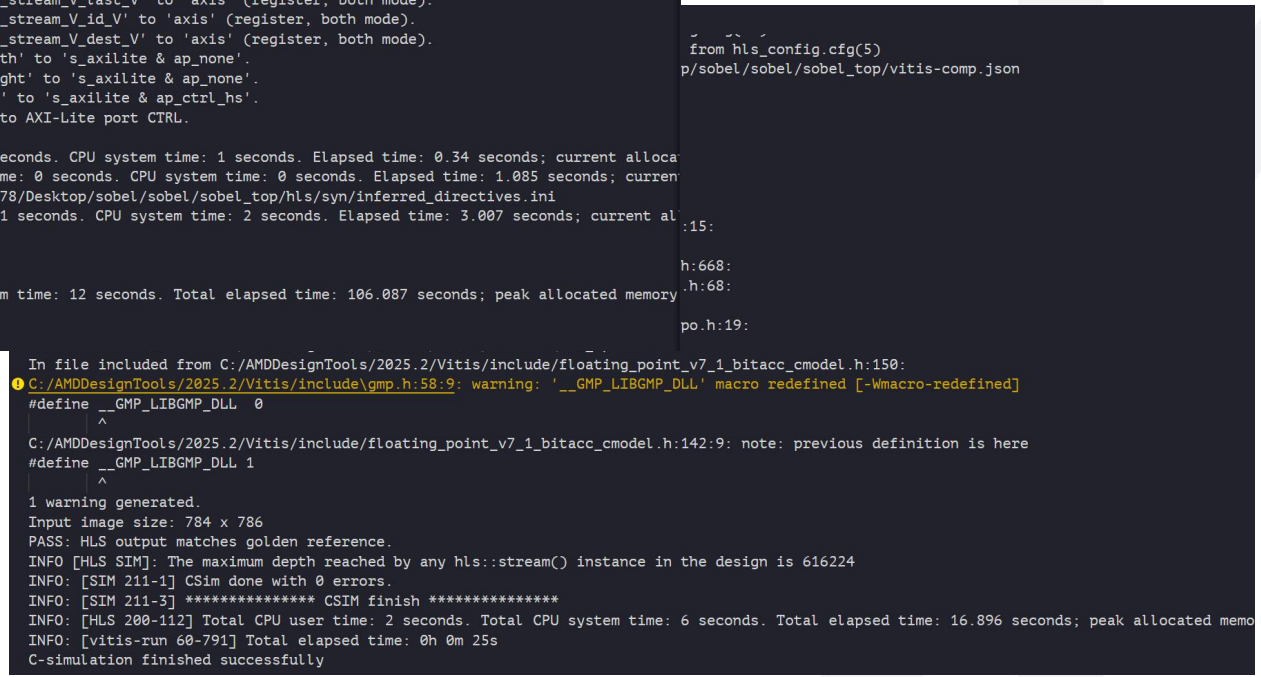
TARGET	WITH UNCERTAINTY	ESTIMATED
8.00 ns	5.84 ns	5.799 ns

Performance & Resource Estimates

Search | Modules | Loops | Hide empty columns

LATENCY(NS)	ITERATION LATENCY	INTERVAL	TRIP COUNT	PIPELINED	STRUCTURE	BRAM	DSP	FF	LUT	URAM
2.951E7		3,688,338		no	function	2	1	1,197	1,839	0
2.951E7		3,688,334		no	function	2	1	1,002	1,572	0
1.538E4		1,921		loop pipeline stpst	function	0	0	25	62	0
1.536E4	2	1	1,920	yes	loop					
2.949E7		3,686,401		loop pipeline stpst	function	0	0	888	1,090	0
2.949E7	8	1	3,686,400	yes	loop					


OUTPUT
PROBLEMS 3
sobel::synthesis
Vitis Messages
sobel::c-simulation
sobel::synthesis
sobel::co-simulation
sobel::package
INFO: [HLS 200-10] -- Generating RTL for module 'sobel_top'
INFO: [HLS 200-10] ----------
WARNING: [RTGEN 206-101] Design contains AXI ports. Reset is fixed to synchronous and active low.
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/in_stream_V_data_V' to 'axis' (register, both mode).
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/in_stream_V_keep_V' to 'axis' (register, both mode).
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/in_stream_V_strb_V' to 'axis' (register, both mode).
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/in_stream_V_user_V' to 'axis' (register, both mode).
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/in_stream_V_last_V' to 'axis' (register, both mode).
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/in_stream_V_id_V' to 'axis' (register, both mode).
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/in_stream_V_dest_V' to 'axis' (register, both mode).
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/out_stream_V_data_V' to 'axis' (register, both mode).
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/out_stream_V_keep_V' to 'axis' (register, both mode).
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/out_stream_V_strb_V' to 'axis' (register, both mode).
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/out_stream_V_user_V' to 'axis' (register, both mode).
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/out_stream_V_last_V' to 'axis' (register, both mode).
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/out_stream_V_id_V' to 'axis' (register, both mode).
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/out_stream_V_dest_V' to 'axis' (register, both mode).
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/width' to 's_axilite & ap_none'.
INFO: [RTGEN 206-500] Setting interface mode on port 'sobel_top/height' to 's_axilite & ap_none'.
INFO: [RTGEN 206-500] Setting interface mode on function 'sobel_top' to 's_axilite & ap_ctrl_hs'.
INFO: [RTGEN 206-100] Bundling port 'width', 'height' and 'return' to AXI-Lite port CTRL.
INFO: [RTGEN 206-100] Finished creating RTL model for 'sobel_top'.
INFO: [HLS 200-111] Finished Creating RTL model: CPU user time: 0 seconds. CPU system time: 1 seconds. Elapsed time: 0.34 seconds; current alloca
INFO: [HLS 200-111] Finished Generating all RTL models: CPU user time: 0 seconds. CPU system time: 0 seconds. Elapsed time: 1.085 seconds; curren
INFO: [HLS 200-2225] Wrote inferred directives to file C:/Users/57778/Desktop/sobel/sobel/sobel_top/hls/syn/inferred_directives.ini
INFO: [HLS 200-111] Finished Updating report files: CPU user time: 1 seconds. CPU system time: 2 seconds. Elapsed time: 3.007 seconds; current al
INFO: [VHDL 208-304] Generating VHDL RTL for sobel_top.
INFO: [VLOG 209-307] Generating Verilog RTL for sobel_top.
INFO: [HLS 200-789] **** Estimated Fmax: 172.44 MHz
INFO: [HLS 200-112] Total CPU user time: 6 seconds. Total CPU system time: 12 seconds. Total elapsed time: 106.087 seconds; peak allocated memory
INFO: [v++ 60-791] Total elapsed time: 0h 1m 54s
Synthesis finished successfully, open report.


from hls_config.cfg(5)
p/sobel/sobel/sobel_top/vitis-comp.json
:15:
h:668:
.h:68:
po.h:19:
In file included from C:/AMDDesignTools/2025.2/Vitis/include/floating_point_v7_1_bitacc_cmodel.h:150:
C:/AMDDesignTools/2025.2/Vitis/include\gmp.h:58:9: warning: '__GMP_LIBGMP_DLL' macro redefined [-Wmacro-redefined]
#define __GMP_LIBGMP_DLL 0
^
C:/AMDDesignTools/2025.2/Vitis/include/floating_point_v7_1_bitacc_cmodel.h:142:9: note: previous definition is here
#define __GMP_LIBGMP_DLL 1
^
1 warning generated.
Input image size: 784 x 786
PASS: HLS output matches golden reference.
INFO [HLS SIM]: The maximum depth reached by any hls::stream() instance in the design is 616224
INFO: [SIM 211-1] CSim done with 0 errors.
INFO: [SIM 211-3] *************** CSIM finish ***************
INFO: [HLS 200-112] Total CPU user time: 2 seconds. Total CPU system time: 6 seconds. Total elapsed time: 16.896 seconds; peak allocated memo
INFO: [vitis-run 60-791] Total elapsed time: 0h 0m 25s
C-simulation finished successfully

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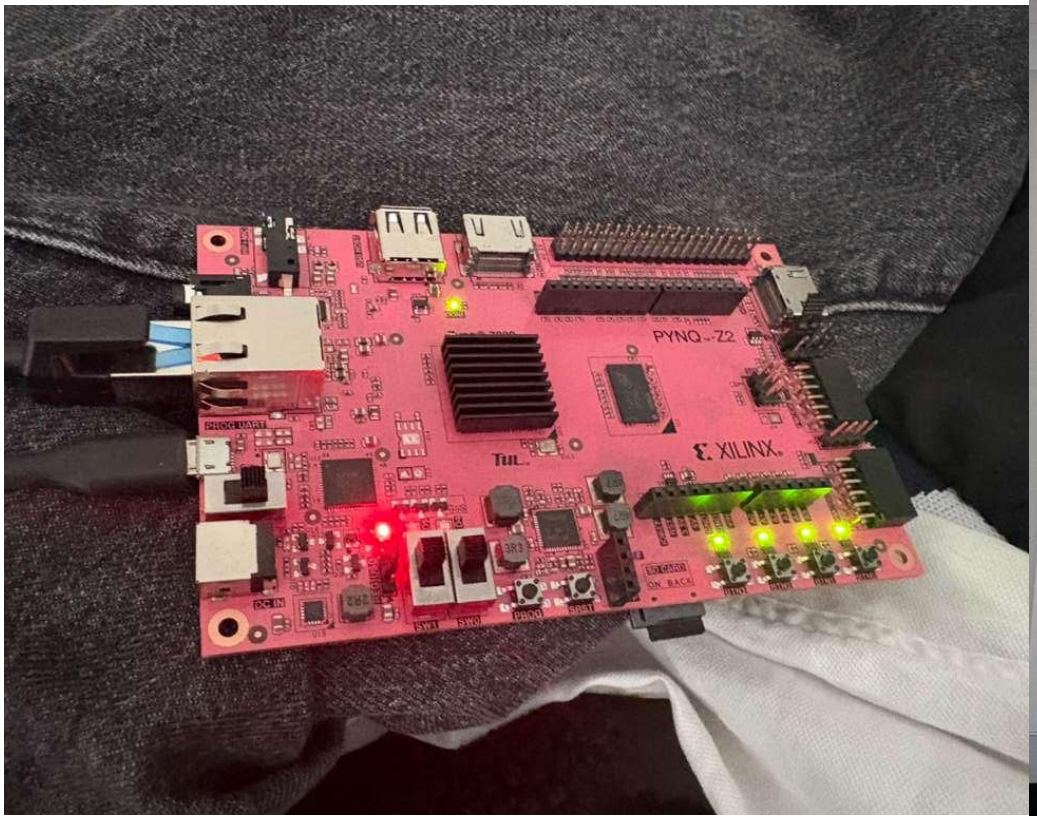
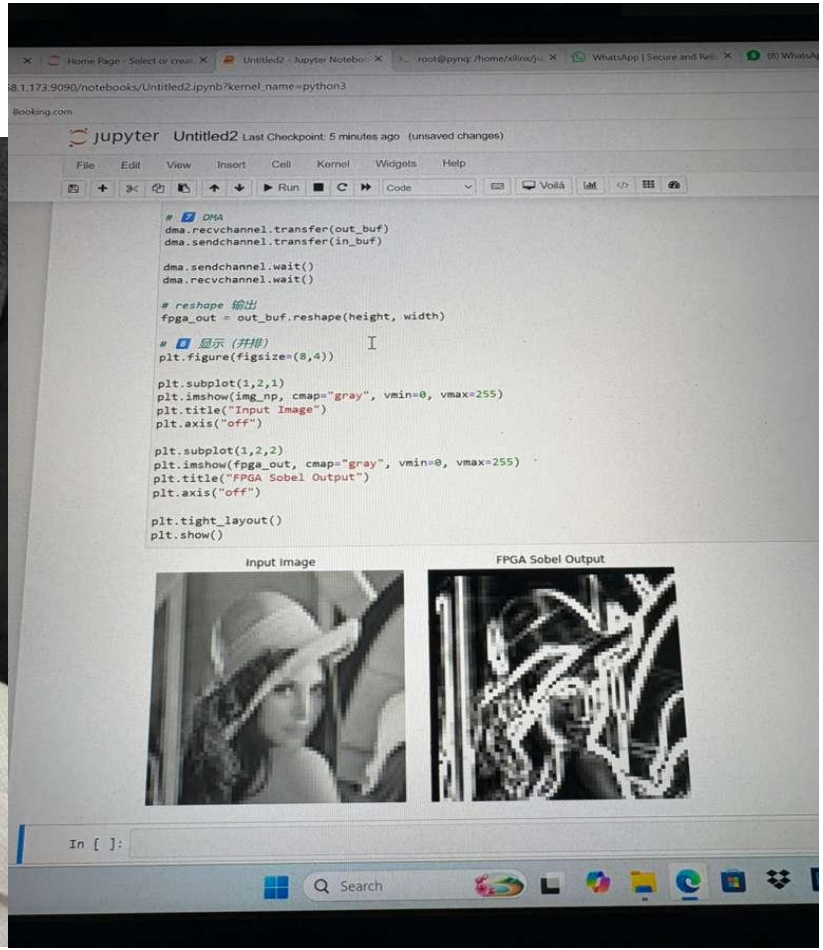
Final deploy





Thanks!

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